I. Fourier components and their harmonic qualities (From Quinn 2007, Yust 2016)

f_0	f_1	f_2	f_3	f_4	f_5	f_6
Cardinality	Chromaticism	Quartal Quality	Hexatonicity/ Tradicity	Octatonicity	Diatonicity	Whole Tone Quality

II. Example collections and their DFTs

[0,2,4,5,7,9] (Diatonic/Guidonian hexachord starting on C)

	f_1	f_2	f_3	f_4	f_5	f_6
Magnitude	1.04	0	1.41	0	3.86	0
Phase	4.5	0	1.5	0	10.5	0

[4,6,8,9,11,1] (Diatonic/Guidonian hexachord starting on E)

Note how the magnitudes are all the same, while the phase only differs for f_1 and f_5. C and E are "hexatonically" related.

	f_1	f_2	f_3	f_4	f_5	f_6
Magnitude	1.04	0	1.41	0	3.86	0
Phase	8.5	0	1.5	0	6.5	0

[0,1,4,5,8,9] (Hexatonic collection starting on C)

The symmetry of the collection balances out in all coefficients except f_3. Note that this collection shares Ph_3 with both of the diatonic collections above.

	f_1	f_2	f_3	f_4	f_5	f_6
Magnitude	0	0	4.24	0	0	0
Phase	0	0	1.5	0	0	0

[0,1,4,6,7,10] (Petrushka chord, C major + F major triads)

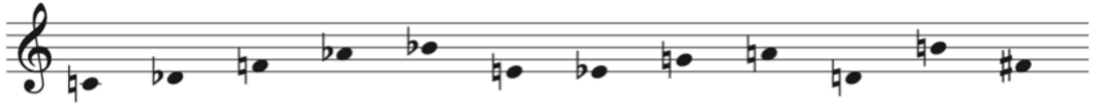
	f_1	f_2	f_3	f_4	f_5	f_6
Magnitude	0	2	0	3.46	0	2
Phase	0	0	0	3	0	0

For more fun, here's a simple calculator (requires Google account):
https://colab.research.google.com/drive/1mqgurRGXXu6PnY8H9Wu8hkS46N9ZbfQS?usp=sharing

Or, how diatonic is my favorite hexachord?
https://docs.google.com/spreadsheets/d/1vCxSmhkHOcABWnjv5n8zAmNJt9ClfZWRTV1qZSBIH8A/edit?usp=sharing

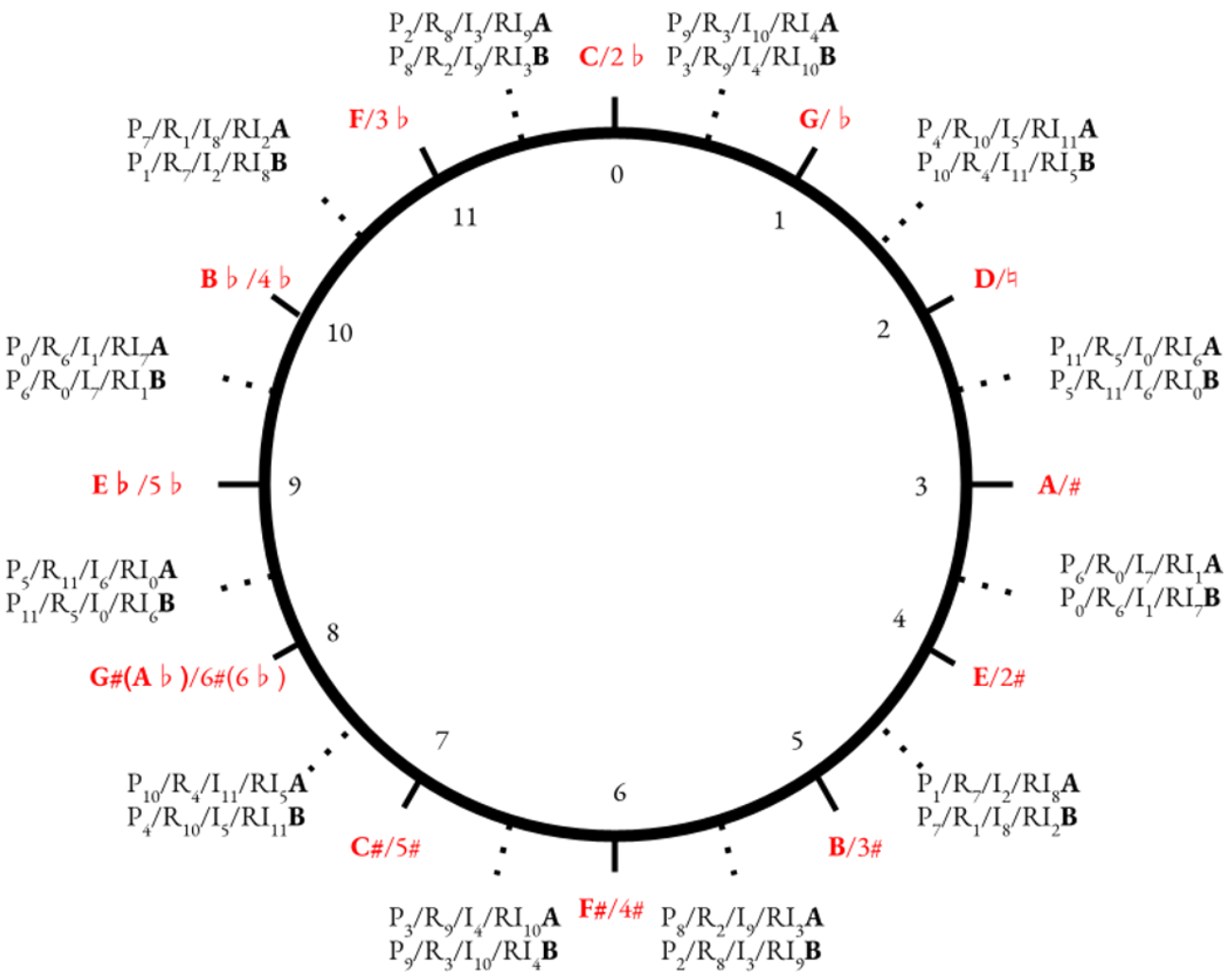
III. Quaderno

Row and DFT:

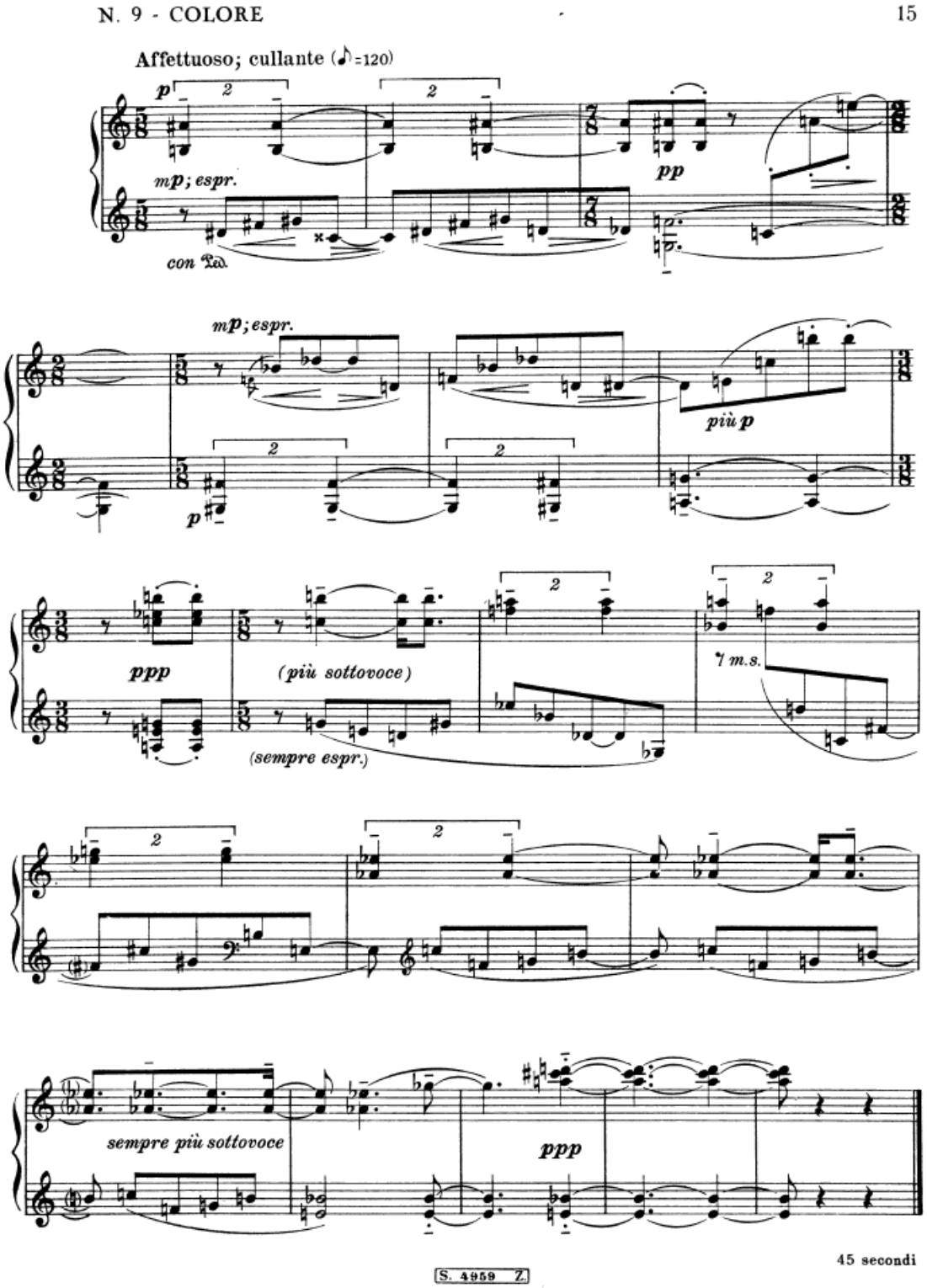


Segment	Set class	$\lvert f_1 \rvert^2$	$\lvert f_2 \rvert^2$	$\lvert f_3 \rvert^2$	$\lvert f_4 \rvert^2$	$\lvert f_5 \rvert^2$	$\lvert f_6 \rvert^2$
Trichord first	(015)	2	4	5	0	2	1
Trichord last	(037)	0.27	1	5	3	3.73	1
Trichords middle	(026)	1	3	1	3	1	9
Tetrachord first	(0148)	0.27	3	8	1	3.73	0
Tetrachord last	(0358)	0.54	0	4	4	7.46	0
Tetrachord middle	(0147)	1	3	4	7	1	0
Pentachords	(01358)	1.54	1	5	1	8.46	1
Hexachords	(014579)	0.27	1	8	3	3.73	4

Ph_5 circle


0
C/2 ♭
$P_2/R_8/I_3/RI_9$**A**
$P_8/R_2/I_9/RI_3$**B**
$P_9/R_3/I_{10}/RI_4$**A**
$P_3/R_9/I_4/RI_{10}$**B**
1
G/ ♭
$P_4/R_{10}/I_5/RI_{11}$**A**
$P_{10}/R_4/I_{11}/RI_5$**B**
2
D/♮
$P_{11}/R_5/I_0/RI_6$**A**
$P_5/R_{11}/I_6/RI_0$**B**
3
A/#
$P_6/R_0/I_7/RI_1$**A**
$P_0/R_6/I_1/RI_7$**B**
4
E/2#
$P_1/R_7/I_2/RI_8$**A**
$P_7/R_1/I_8/RI_2$**B**
5
B/3#
$P_8/R_2/I_9/RI_3$**A**
$P_2/R_8/I_3/RI_9$**B**
6
F#/4#
$P_3/R_9/I_4/RI_{10}$**A**
$P_9/R_3/I_{10}/RI_4$**B**
7
C#/5#
$P_{10}/R_4/I_{11}/RI_5$**A**
$P_4/R_{10}/I_5/RI_{11}$**B**
8
G#(A ♭)/6#(6 ♭)
$P_5/R_{11}/I_6/RI_0$**A**
$P_{11}/R_5/I_0/RI_6$**B**
9
E ♭ /5 ♭
$P_0/R_6/I_1/RI_7$**A**
$P_6/R_0/I_7/RI_1$**B**
10
B ♭ /4 ♭
$P_7/R_1/I_8/RI_2$**A**
$P_1/R_7/I_2/RI_8$**B**
11
F/3 ♭
$P_9/R_3/I_{10}/RI_4$**A**
$P_3/R_9/I_4/RI_{10}$**B**

Scores:


N. 9 - COLORE
15
Affettuoso; cullante (♪=120)
p
mp; espr.
con Ped.
pp
mp; espr.
più p
p
ppp
(più sottovoce)
(sempre espr.)
7 m.s.
sempre più sottovoce
ppp
45 secondi
S. 4959 Z.

N. 11 - QUARTINA

Molto lento; fantastico (♩= 40)

p

con grande espress.

pp *sost.*

espr.

non arp.

con la massima espressione

(sostenutiss.)

espr.

m. s.

ppp

ppp

1 min. 20 secondi
Montréal - New York - Mexico City
(Agosto - Settembre 1952)
[ritoccato nell'estate 1953]

S. 4959 Z.

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Appendix (the math):

To do a DFT on pitch class sets (12-EDO):

1. Take a pitch class set, for example:

   [C,D,E,F,G,A]
   [0,2,4,5,7,9]

2. Convert it into a one-hot encoding, that is, first put down a vector with 12 elements:

   [0,0,0,0,0,0,0,0,0,0,0,0]

   For every pitch class *x* in the set, turn the *x*th element of this vector into a 1:

   C D E F G A
   [1,0,1,0,1,1,0,1,0,1,0,0]

   This is called the **characteristic function** of the pc set.

3. Basic equation for DFT on pitch class sets:

$$f_k = \sum_{n=0}^{11} x_n e^{\frac{-i2\pi}{12}nk}$$

   Breaking down the equation:

$$f_k = \sum_{n=0}^{11} x_n e^{\frac{-i2\pi}{12}nk}$$

   f_k Fourier Coefficient: 7 complex numbers

   $\sum_{n=0}^{11}$ Summation Function: add up all the numbers

   x_n Pitch Weight: the number at the *n*th location at our characteristic function

   $e^{\frac{-i2\pi}{12}nk}$ **Mathematical shorthand** that fits the function to various periodicities

   Euler's formula: $e^{ix} = \cos x + i \sin x$

$$e^{\frac{-i2\pi}{12}nk} = \cos\frac{-2\pi nk}{12} + i \sin\frac{-2\pi nk}{12}$$

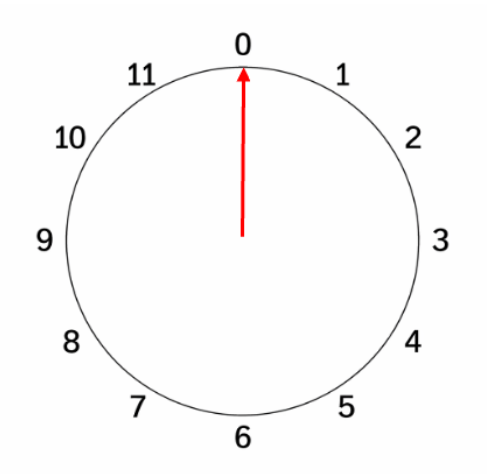
4. Put in the numbers. For the first coefficient, k=1, so:

$$f_1 = \sum_{n=0}^{11} x_n e^{\frac{-i2\pi}{12}n1}$$

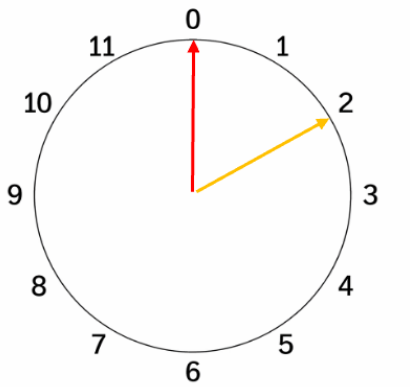
5. We will be adding 12 components, with n going from 0 to 11. In other words, the equation is:

$$f_k = x_0 e^{\frac{-i2\pi}{12}0} + x_1 e^{\frac{-i2\pi}{12}1} + \ldots + x_{11} e^{\frac{-i2\pi}{12}11}$$

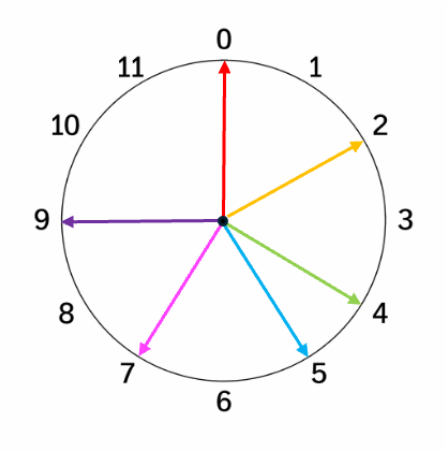
Every e term points the summand in a certain direction. For example, the first term reduces to $e^0 = 1$, which will point it directly East at the positive X axis (Note that with this equation, the angles are tilted by 90°. I rotated the phase circles to better match the traditional pitch class clock, with 0 at the top). If there is a numerical value at x_0 (in this case, 1), then the first summand puts down a vector of length one in the direction specified by the e term:



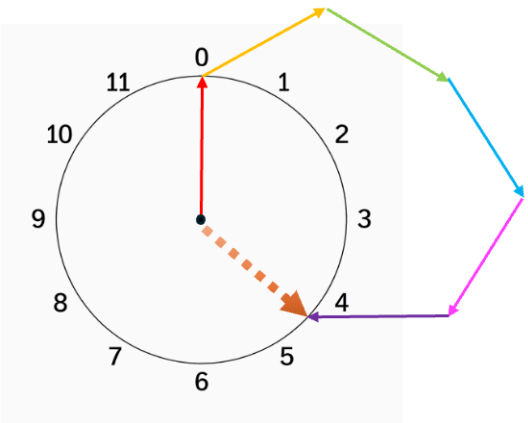
Iterate this for every summand. For the the second term, since there is a zero (C#), there is no vector at all and so we skip it. The next vector to put down is x_3, D:



Etc, until every summand is accounted for:

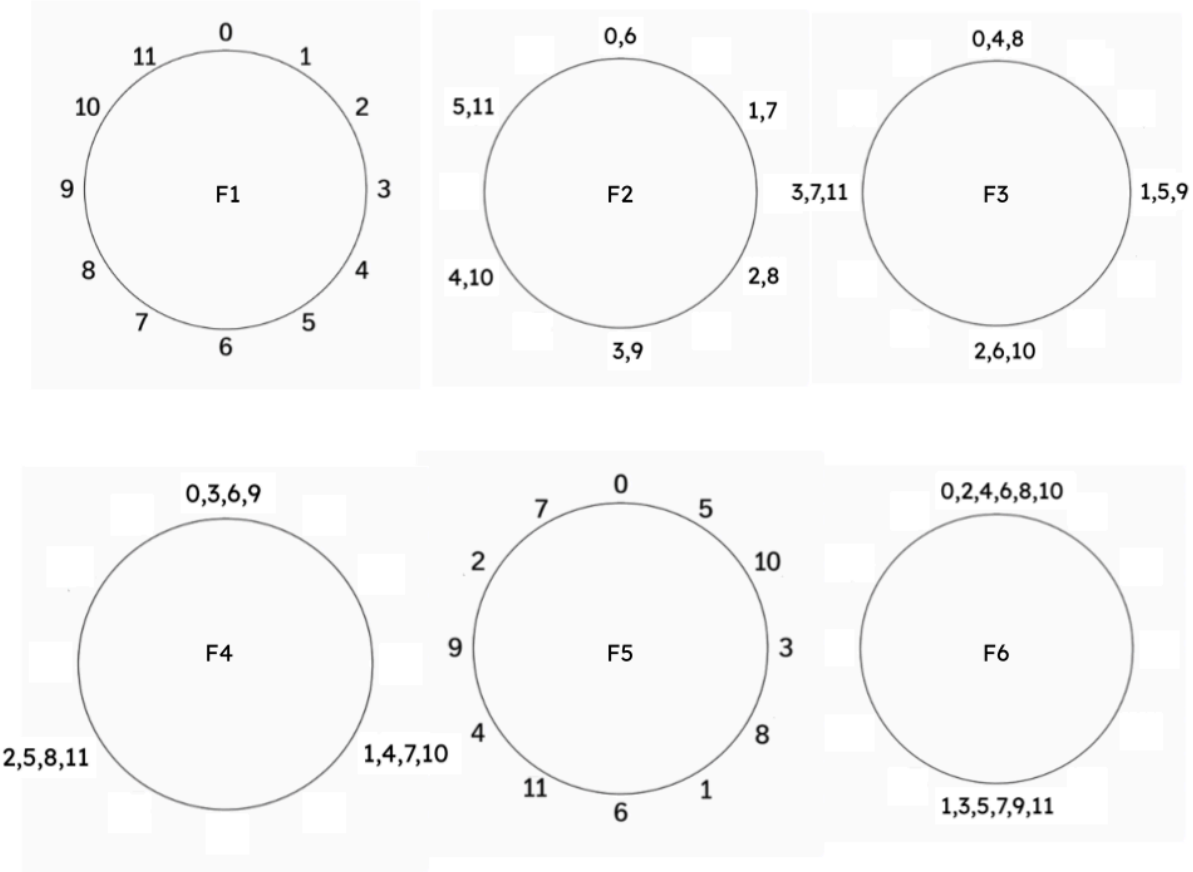
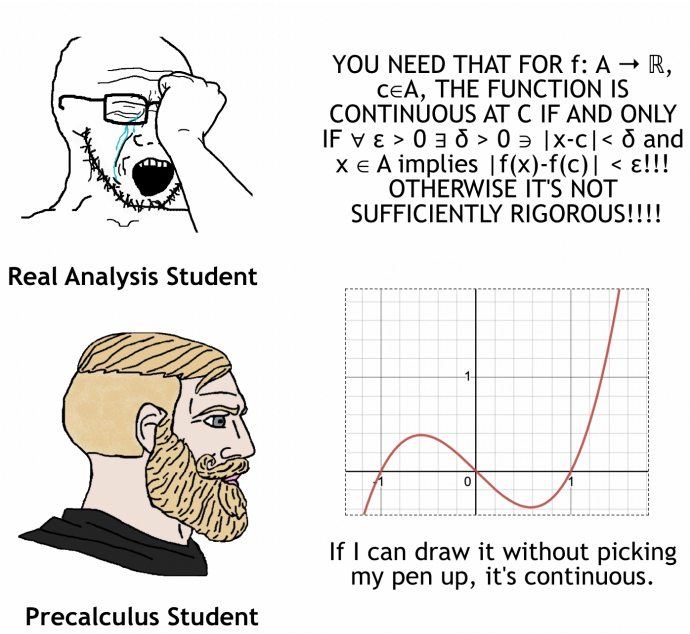


Mathematically, vector addition here is done by utilizing euler's identity to break the e term into cos(x) and isin(x), then adding them componentwise. For those who enjoy by-hand calculations, trig formulas find their uses here. What this does, though, is essentially vector addition, the process of chaining all our arrows without stretching or rotating them (only translation):



The final vector, drawn from the origin to the end point of the last arrow, is the DFT. Note how some pitches will "undo" the force exerted by other pitches. Thus, chromatic clusters under the DFT will show high values for f_1.

The other components change the value k in the equation, which multiplies the *angle* that the e term points towards. In effect, this will change the circle in the following ways:



It's a fun exercise to print these out and start adding vectors by hand, comparing how they behave differently on the different fourier circles – much more fun than having to actually calculate the terms by hand. Or don't do any of this and play around with the calculators and tools out there already: